



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date:10/08/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3 rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical - 3 Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

PROBLEM STATEMENT: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

Objective: To design and develop responsive web pages for **Home, About, Registration, Dashboard, and Events** using **CSS Grid, Flexbox, and optionally Bootstrap/Tailwind CSS**, ensuring that the layouts adapt effectively to different screen sizes and devices following a **mobile-first approach**.

Code :

```
:root {
  --color-bg: #f7f5ee;
  --color-surface: #ffffff;
  --color-ink: #21262b;
  --color-muted: #5b6470;
  --color-primary: #1e3a5f;
  --color-primary-dark: #142a45;
  --color-accent: #a9843f;
  --color-border: #ddd6c6;

  --font-display: "Source Serif 4", Georgia, "Times New Roman", serif;
  --font-body: "Inter", -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
  --font-mono: ui-monospace, "SFMono-Regular", Menlo, monospace;
```

```
--space-1: 0.5rem;
--space-2: 1rem;
--space-3: 1.5rem;
--space-4: 2.5rem;

--radius: 4px;
--max-width: 72rem;
}

/* ---- Reset & base ---- */
* { box-sizing: border-box; }

body {
  margin: 0;
  background: var(--color-bg);
  color: var(--color-ink);
  font-family: var(--font-body);
  line-height: 1.6;
  font-size: 1rem;
}

h1, h2, h3 {
  font-family: var(--font-display);
  color: var(--color-primary-dark);
  line-height: 1.2;
  margin: 0 0 var(--space-2);
}

h1 { font-size: 1.75rem; }
h2 { font-size: 1.3rem; }
h3 { font-size: 1.05rem; }

p { margin: 0 0 var(--space-2); }

a {
  color: var(--color-primary);
  text-decoration-thickness: 1px;
}

a:hover { color: var(--color-accent); }

img { max-width: 100%; display: block; }

a:focus-visible,
button:focus-visible,
input:focus-visible,
textarea:focus-visible {
  outline: 3px solid var(--color-accent);
  outline-offset: 2px;
}

/* =====
```

```
header {
  background: var(--color-primary);
  color: #fff;
  padding: var(--space-2);
  display: flex;
  flex-direction: column;
  gap: var(--space-2);
}

header h1 { color: #fff; font-size: 1.4rem; margin: 0; }

header img { height: 2.5rem; border-radius: var(--radius); }

.brand {
  display: flex;
  align-items: center;
  gap: var(--space-1);
}

header nav ul {
  list-style: none;
  display: flex;
  flex-direction: column;
  gap: var(--space-1);
  padding: 0;
  margin: 0;
}

header nav a {
  color: #fff;
  display: block;
  padding: var(--space-1);
  border-radius: var(--radius);
  text-decoration: none;
}

header nav a[aria-current="page"] {
  background: rgba(255, 255, 255, 0.18);
  font-weight: 600;
}

header nav a:hover { color: var(--color-accent); }

@media (min-width: 40em) {
  header {
    flex-direction: row;
    align-items: center;
    justify-content: space-between;
    padding: var(--space-2) var(--space-3);
  }

  header nav ul {
    flex-direction: row;
    flex-wrap: wrap;
  }
}
```

```
}  
}  
  
#main-content {  
  display: grid;  
  grid-template-columns: 1fr;  
  gap: var(--space-3);  
  max-width: var(--max-width);  
  margin: 0 auto;  
  padding: var(--space-3) var(--space-2);  
}  
  
#main-content > h1 { grid-column: 1 / -1; }  
  
@media (min-width: 48em) {  
  #main-content:has(> aside) {  
    grid-template-columns: 2fr 1fr;  
    align-items: start;  
  }  
  
  #main-content:has(> aside) > h1 {  
    grid-column: 1 / -1;  
  }  
  
  #main-content:has(> aside) > aside {  
    grid-column: 2;  
  }  
}  
  
section { min-width: 0; }  
  
article {  
  background: var(--color-surface);  
  border: 1px solid var(--color-border);  
  border-top: 4px solid var(--color-primary);  
  border-radius: var(--radius);  
  padding: var(--space-2);  
}  
  
article h3 { margin-top: 0; }  
  
section:has(> article) {  
  display: grid;  
  gap: var(--space-2);  
  grid-template-columns: repeat(auto-fit, minmax(15rem, 1fr));  
}  
  
aside {  
  background: var(--color-surface);  
  border: 1px solid var(--color-border);  
  border-left: 4px solid var(--color-accent);  
  border-radius: var(--radius);
```

```
padding: var(--space-2);
}

aside ul { padding-left: 1.1rem; margin: 0; }
aside li { margin-bottom: var(--space-1); }

form {
  display: flex;
  flex-direction: column;
  gap: var(--space-2);
  max-width: 26rem;
}

form label {
  font-weight: 600;
  font-size: 0.9rem;
}

form br { display: none; }

input, textarea {
  font-family: var(--font-body);
  font-size: 1rem;
  padding: var(--space-1);
  border: 1px solid var(--color-border);
  border-radius: var(--radius);
  width: 100%;
}

textarea { resize: vertical; }

button {
  font-family: var(--font-body);
  background: var(--color-primary);
  color: #fff;
  border: none;
  padding: var(--space-1) var(--space-3);
  border-radius: var(--radius);
  cursor: pointer;
  font-size: 1rem;
  align-self: flex-start;
}

button:hover { background: var(--color-primary-dark); }

table {
  border-collapse: collapse;
  width: 100%;
  min-width: 32rem;
  background: var(--color-surface);
  display: block;
```

```
    overflow-x: auto;
}

th, td {
    border: 1px solid var(--color-border);
    padding: var(--space-1) var(--space-2);
    text-align: left;
}

th { background: var(--color-bg); font-family: var(--font-mono); font-size:
0.85rem; }

caption {
    text-align: left;
    font-weight: 600;
    margin-bottom: var(--space-1);
}

footer {
    border-top: 1px solid var(--color-border);
    padding: var(--space-3) var(--space-2);
    text-align: center;
    color: var(--color-muted);
    font-size: 0.9rem;
}

footer nav {
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: var(--space-1);
    margin-bottom: var(--space-1);
}

footer nav a { color: var(--color-muted); }
footer nav a:hover { color: var(--color-accent); }
```

Output :

Login.html:-

CHARUSAT **login**

[Home](#) [About](#) [Events](#) [FAQ](#) [Contact](#) [Feedback](#) [Login](#) [Register](#)

Login to Your Account

Username

Password

Login

[Forgot Password?](#)

Don't have an account? [Register](#)

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Home.html:-

CHARUSAT **StudentHub**

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Welcome to StudentHub

Track attendance, manage assignments, stay updated on campus events, and share feedback — all from a single, easy-to-use dashboard built for CHARUSAT students.

[Login](#) [Register](#)

Latest Announcements

Semester Exam Schedule Released

The examination timetable for the upcoming semester has been published. Check your department notice board or the Events page for exact dates and venues.

[Read more](#)

Annual Tech Fest Registrations Open

Sign up now for workshops, coding contests, and guest talks at this year's tech fest. Limited seats available for select sessions.

[Read more](#)

Library Timings Extended

The central library will now stay open until 9 PM on weekdays during the exam period to support student preparation.

[Read more](#)

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Events.html:-

CHARUSAT logo

Upcoming Events

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FAQ

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Login

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All Events

Annual Tech Fest

Date: March 15, 2026

Venue: Main Auditorium

[Register]

Blood Donation Camp

Date: April 2, 2026

Venue: Sports Complex Ground

[Register]

Event Categories

• Technical

• Cultural & Social

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Dashboard.html:-

CHARUSAT logo

Dashboard

Dashboard

Profile

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Logout

Welcome, Student!

Attendance

You have attended 42 out of 45 classes this semester — an overall attendance of 93%.

Assignments

You have 2 pending assignments: Data Structures Lab Report (due Friday) and Database Systems Case Study (due next Monday).

Events

Upcoming Event

Annual Tech Fest — March 15, 2026 at the Main Auditorium. Registration closes March 10.

Notices

Mid-semester exams begin October 10. Check the Events page for your full timetable.

Recent Activity

Submitted Assignment 3 for Data Structures.

Registered for the Annual Tech Fest.

Marked present for today's Database Systems lecture.

Contact | About

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ContactUs.html:-

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Have a Question?

Visit our [FAQ page](#) or submit the [feedback form](#).

Office Hours

Monday – Friday: 9:00 AM to 5:00 PM
Saturday: 10:00 AM to 1:00 PM
Sunday: Closed

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Learning outcome : After completing this practical, I will be able to understand and implement the concepts of **responsive web design** using CSS Grid and Flexbox. I will be able to create responsive layouts for different web pages such as **Home, About, Registration, Dashboard, and Events**, and make them adaptable to different screen sizes using a **mobile-first approach**. I will also be able to use **Bootstrap/Tailwind CSS** for designing responsive and user-friendly interfaces.